

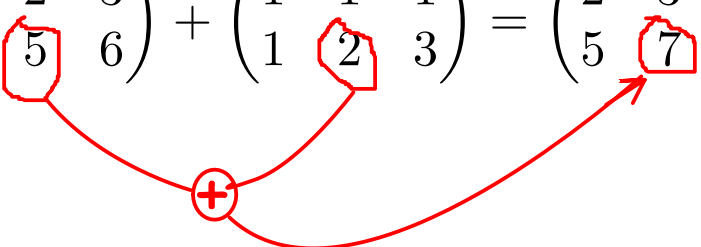
The algebra of matrices

- add matrices
- subtract matrices
- multiply a matrix by a number
- multiply two matrices
- divide matrices (sort of, sometimes)

Adding two matrices

We can add matrices if they have the same size, i.e., if they have the same number of rows and columns.

Example

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} + \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{pmatrix} = \begin{pmatrix} 2 & 3 & 4 \\ 5 & 7 & 9 \end{pmatrix}$$


For all i, j , $(A + B)_{ij} = A_{ij} + B_{ij}$

Example

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} + \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \quad \text{is undefined}$$

not the same size

Commutative:

For all matrices A, B ,

$$A + B = B + A$$

Associative:

For all matrices A, B, C ,

$$(A + B) + C = A + (B + C)$$

Follows directly from commutativity and associativity of addition of real (and complex) numbers.

(see Exercise 1 in Notes)

Fact For every matrix A ,

$$A + \mathbf{0} = \mathbf{0} + A = A.$$


zero matrix of the
same size as A

$\mathbf{0}$ is an additive identity.

Example

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} + \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$$

Notation: $A = (a_{ij})$

Definition (negative of a matrix)

$$-A = (-a_{ij})$$

If $A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$ then $-A = \begin{pmatrix} -a_{11} & -a_{12} \\ -a_{21} & -a_{22} \end{pmatrix}$.

Example $-\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} = \begin{pmatrix} -1 & -2 & -3 \\ -4 & -5 & -6 \end{pmatrix}$

Fact For any matrix A ,

$$A + (-A) = (-A) + A = \mathbf{0}.$$

$-A$ is the additive inverse of A .

Subtracting two matrices

Definition For any matrices A, B

$$A - B = A + (-B).$$

Example

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} - \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 2 & 3 \end{pmatrix}$$

Scalar multiplication

"scalar" = real or complex number
(usually a real number in this module)

We can multiply a matrix by a scalar:

Example Suppose $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 2 & 1 \end{pmatrix}$.

$$\text{Then } 2A = \begin{pmatrix} 2 & 4 \\ 6 & 8 \\ 4 & 2 \end{pmatrix}.$$

Definition

Let $A = (a_{ij})$ be a matrix and r a scalar.

Then $r \cdot A$ is the matrix s.t. for all i, j ,

$$(r \cdot A)_{ij} = r \cdot a_{ij}.$$

Fact Let A, B matrices of the same size and s, t scalars.

Then the following hold:

1.) $(s + t)A = sA + tA,$

2.) $s(A + B) = sA + sB,$

3.) $(st)A = s(tA),$

For example,

$$15 \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = 5 \underbrace{\left(3 \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \right)}_{\begin{pmatrix} 3 & 6 \\ 9 & 12 \end{pmatrix}}$$

4.) $1 \cdot A = A$

5.) $(-1) \cdot A = -A.$

The transpose of a matrix

Definition

Given an $m \times n$ matrix A , the transpose of A is the $n \times m$ matrix A^T whose columns are formed by corresponding rows of A :

$$(A^T)_{ij} = A_{ji}$$

Example

If $B = \begin{pmatrix} 1 & -1 & 2 \\ 3 & 4 & 5 \end{pmatrix}$ then $B^T = \begin{pmatrix} 1 & 3 \\ -1 & 4 \\ 2 & 5 \end{pmatrix}$

Definition

A matrix A is symmetric if $A = A^T$, i.e., if A is square and for all i, j ,

$$a_{ij} = a_{ji}.$$

Example

$$\begin{pmatrix} 1 & -1 & 3 \\ -1 & 7 & 4 \\ 3 & 4 & 5 \end{pmatrix} \text{ is symmetric}$$

$$\begin{pmatrix} 1 & -1 & \underline{8} \\ -1 & 7 & 4 \\ \underline{3} & 4 & 5 \end{pmatrix} \text{ is not symmetric}$$

- Fact 1.) $(A^T)^T = A$,
- 2.) $(A + B)^T = A^T + B^T$,
- 3.) for any scalar r ,
- $$(rA)^T = rA^T.$$